

# OPERATING SYSTEMS EXERCISE 1

## Assignment A – binary-digits

Write a C-program that visualizes data in its binary representation. In other words, by printing *1*s and *0*s.

### SYNOPSIS:

```
binary-digits [-d DELAY] [-o OUTPUTFILE] [FILE]...
```

The program shall read each file given as positional argument (or `stdin` if there are no positional arguments) and print it byte per byte to `stdout` or a specified output file in **big endian byte order**.<sup>1</sup>

Your program must work on a stream of characters. I.e. not wait until all input is available, but process the data immediately.

The program must run until either all files specified as positional arguments are processed, there is no more input from `stdin` (i.e. *EOF* is reached) or the program receives some signal demanding termination, whichever comes first. No special signal handling is necessary.

For example a call to the program with a file as an argument only containing the ASCII characters `a`, `b` and a new-line character (`\n`), should produce this output:

```
011000010110001000001010
```

3 \* 8 digits, because every character is represented by 8 bits.

If the option `-o` is given, the output is written to the specified file `outfile`. Otherwise, the output is written to `stdout`.

If the option `-d` is given, the program should wait the specified amount of seconds after each digit written. The delay may be specified as an ASCII represented decimal floating point number, e.g. `0.5`.

### Hints

- The functions `nanosleep(2)` and `strtod(3)` might be helpful for implementing the delay part.
- `fgetc(3)` and `fputc(3)` can be used to operate on character streams.
- It might be necessary to `fflush(3)` a stream in order to generate steady output.

### Testing

Test your program with various inputs, such as a file `in1.txt` containing `"OSUE\n"`.

Executing your program should produce:

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<sup>1</sup><https://en.wikipedia.org/wiki/Endianness>

```
$ ./binary-digits in1.txt  
0100111101010011010101010100010100001010
```

Or you can let the program read from `stdin` interactively, by just calling

```
$ ./binary-digits
```

and then entering arbitrary letters on your keyboard. This way you can see the representation for anything you want. Note that the shell input is buffered and you will only see a result every time a new-line character (`\n`) is sent by pressing return. End the input with `Ctrl+D`.

A more complex invocation could be:

```
$ ./binary-digits -o outfile.txt -d 0.5 in1.txt in2.txt
```

This will store the contents of `in1.txt` and `in2.txt` expanded to 1 and 0 characters in `outfile.txt`, but wait 0.5 seconds after writing each character.

What you could also do is to read random input from `/dev/urandom` and create an endless stream of text.

```
$ ./binary-digits -d 0.05 /dev/urandom
```

# Coding Rules and Guidelines

Your score depends upon the compliance of your submission to the presented guidelines and rules. Violations result in deductions of points. Hence, before submitting your solution, go through the following list and check if your program complies.

## Rules

Compliance with these rules is essential to get any points for your submission. A violation of any of the following rules results in 0 points for your submission.

1. All source files of your program(s) must compile via

```
$ gcc -std=c99 -pedantic -Wall -D_DEFAULT_SOURCE -D_BSD_SOURCE -D_SVID_SOURCE  
-D_POSIX_C_SOURCE=200809L -g -c filename.c
```

without *errors* and your program(s) must link without *errors*. The compilation flags must be used in the Makefile. The feature test macros must not be bypassed (i.e., by undefining these macros or adding some in the C source code).

2. The functionality of the program(s) must conform to the assignment. The program(s) shall operate according to the specification/assignment given the test cases in the respective assignment.

## General Guidelines

Violation of following guidelines leads to a deduction of points.

1. All source files of your program(s) must compile with

```
$ gcc -std=c99 -pedantic -Wall -D_DEFAULT_SOURCE -D_BSD_SOURCE -D_SVID_SOURCE  
-D_POSIX_C_SOURCE=200809L -g -c filename.c
```

without *warnings and info messages* and your program(s) must link without warnings.

2. There must be a Makefile implementing the targets: **all** to build the program(s) (i.e. generate executables) from the sources (this must be the first target in the Makefile); **clean** to delete all files that can be built from your sources with the Makefile.
3. The program shall operate according to the specification/assignment without major issues (e.g., segmentation fault, memory corruption).
4. Arguments have to be parsed according to UNIX conventions (we strongly encourage the use of **getopt(3)**). The program has to conform to the given synopsis/usage in the assignment. If the synopsis is violated (e.g., unspecified options or too many arguments), the program has to terminate with the usage message containing the program name and the correct calling syntax. Argument handling should also be implemented for programs without arguments.
5. Correct (=normal) termination, including a cleanup of resources.
6. Upon success the program has to terminate with exit code 0, in case of errors with an exit code greater than 0. We recommend to use the macros **EXIT\_SUCCESS** and **EXIT\_FAILURE** (defined in **stdlib.h**) to enable portability of the program.
7. If a function indicates an error with its return value, it *should* be checked in general. If the subsequent code depends on the successful execution of a function (e.g. resource allocation), then the return value *must* be checked.

8. Functions that do not take any parameters have to be declared with **void** in the signature, e.g., `int get_random_int(void);`.
9. Procedures (i.e., functions that do not return a value) have to be declared as **void**.
10. Error messages shall be written to **stderr** and should contain the program name **argv[0]**.
11. It is forbidden to use the functions: **gets**, **scanf**, **fscanf**, **atoi** and **atol** to avoid crashes due to invalid inputs.

| FORBIDDEN           | USE INSTEAD                              |
|---------------------|------------------------------------------|
| <code>gets</code>   | <code>fgets</code>                       |
| <code>scanf</code>  | <code>fgets</code> , <code>sscanf</code> |
| <code>fscanf</code> | <code>fgets</code> , <code>sscanf</code> |
| <code>atoi</code>   | <code>strtol</code>                      |
| <code>atol</code>   | <code>strtol</code>                      |

12. Documentation is mandatory. Format the documentation in Doxygen style (see Wiki and Doxygen's intro).
13. Write meaningful comments. For example, meaningful comments describe the algorithm, or why a particular solution has been chosen, if there seems to be an easier solution at a first glance. Avoid comments that just repeat the code itself (e.g., `i = i + 1; /* i is incremented by one */`).
14. The documentation of a module must include: name of the module, name and student id of the author (**@author** tag), purpose of the module (**@brief**, **@details** tags) and creation date of the module (**@date** tag).  
Also the Makefile has to include a header, with author and program name at least.
15. Each function shall be documented either before the declaration or the implementation. It should include purpose (**@brief**, **@details** tags), description of parameters and return value (**@param**, **@return** tags) and description of global variables the function uses (**@details** tag).  
You should also document **static** functions (see **EXTRACT\_STATIC** in the file **Doxyfile**). Document visible/exported functions in the header file and local (**static**) functions in the C file. Document variables, constants and types (especially **structs**) too.
16. Documentation, names of variables and constants shall be in English.
17. Internal functions shall be marked with the **static** qualifier and are not allowed to be exported (e.g., in a header file). Only functions that are used by other modules shall be declared in the header file.
18. All exercises shall be solved with functions of the C standard library. If a required function is not available in the standard library, you can use other (external) functions too. Avoid reinventing the wheel (e.g., re-implementation of **strcmp**).
19. Name of constants shall be written in upper case, names of variables in lower case (maybe with first letter capital).
20. Use meaningful variable and constant names (e.g., also semaphores and shared memories).
21. Avoid using global variables as far as possible.
22. All boundaries shall be defined as constants (macros). Avoid arbitrary boundaries. If boundaries are necessary, treat its crossing.
23. Avoid side effects with **&&** and **||**, e.g., write `if (b != 0) c = a/b;` instead of `if (b != 0 && c = a/b).`

24. Each **switch** block must contain a **default** case. If the case is not reachable, write **assert(0)** to this case (defensive programming).
25. Logical values shall be treated with logical operators, numerical values with arithmetic operators (e.g., test 2 strings for equality by `strcmp(...) == 0` instead of `!strcmp(...)`).
26. Indent your source code consistently (there are tools for that purpose, e.g., **indent**).
27. Avoid tricky arithmetic statements. Programs are written once, but read more times. Your program is not better if it is shorter!
28. For all I/O operations (read/write from/to **stdin**, **stdout**, files, sockets, pipes, etc.) use *either* standard I/O functions (**fdopen(3)**, **fopen(3)**, **fgets(3)**, etc.) *or* POSIX functions (**open(2)**, **read(2)**, **write(2)**, etc.). Remember, standard I/O functions are buffered. Mixing standard I/O functions and POSIX functions to access a common file descriptor can lead to undefined behaviour and is therefore forbidden.
29. If asked in the assignment, you must implement signal handling (**SIGINT**, **SIGTERM**). You must only use *async-signal-safe* functions in your signal handlers.
30. Close files, free dynamically allocated memory, and remove resources after usage.
31. Don't waste resources due to inconvenient programming. Header files shall not include implementation parts (exception: macros).

## Exercise 1 Guidelines

Violation of following guidelines leads to a deduction of points in exercise 1.

1. Correct use of named semaphores (**sem\_open(3)**, **sem\_close(3)** **sem\_unlink(3)**) and POSIX shared memory (**shm\_overview(7)**) for inter-process communication of separated programs (e.g., server and client).  
Use your matriculation number as prefix in the names of all resources.
2. "Busy waiting" is forbidden. (Busy waiting is the repeated check of a condition in a loop for synchronization purposes.)
3. Synchronization with **sleep** is forbidden.